SUPPORTING ADULT GUIDANCE

Expressions, equations, formulae and identities

Algebraic Expressions and Identities · Lesson 1. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

This lesson opens the unit by separating the four kinds of algebraic statement and introducing ≡. In the main activity, learners slide x through many values and see both sides as numbers and as graphs; an identity's graphs coincide. They then decide identity or equation, and either prove it by expanding or give a counterexample. They sort statements, hunt counterexamples to tempting false claims, and find missing numbers that make an identity. A stretch explains why testing cannot prove an identity.

Driving Question: *Is it true for some values, or for every value?*

The mathematics, in plain terms

An identity holds for all values of the variable; an equation holds for some (possibly none). Finitely many agreeing values never prove an identity — two different polynomials can agree at several points, for example $x(x-1)(x-2)(x-3)+2x$ and $2x$ agree at 0, 1, 2 and 3. A single value where the sides differ disproves an identity. Algebraic manipulation (expanding, simplifying) proves it, because each step is valid for every number. Comparing coefficients finds unknown constants in an identity.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — Test, then prove

Is $2(x+3)=2x+6$ an identity?

1. Test $x=1$: $2\times4=8$, and $2+6=8$.	they agree
2. Test $x=-2$: $2\times1=2$, and $-4+6=2$.	they agree
3. Testing suggests an identity, but cannot check every number.	so prove it
4. Expand: $2\times x+2\times3$.	$=2x+6$

$2(x+3)\equiv2x+6$: proved for every value of x.

Example 2 — A counterexample

Is $(x + 1)^2 = x^2 + 1$ an identity?

1. Test $x = 0$: 1 and 1.	they agree
2. Test $x = 1$: $(1 + 1)^2 = 4$, but $1 + 1 = 2$.	they differ
3. One value where they differ is enough.	not an identity
4. Expanding shows why: $(x + 1)^2 = x^2 + 2x + 1$.	the $2x$ is missing

Not an identity. It is an equation, true only when $x = 0$.

Testing suggests, algebra proves

Learners first gather evidence with the tester — values and graphs — so the difference between 'some' and 'every' is felt before it is named. Every identity is then proved by expanding, and every non-identity is sunk by a counterexample. The stretch makes the logic explicit with an example where four agreeing values mislead. Year 9 met the four kinds of statement; this lesson uses them only as a starting point for ≡ and proof.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	The identity tester: values and graphs, then prove or find a counterexample. Stretch: why testing cannot prove.	~16 min
Movement	Four kinds of statement: expression, equation, formula, identity.	~6 min
Reflection	The counterexample hunt: break tempting false claims; recognise true ones.	~9 min
Creativity	The identity builder: find the missing numbers by matching terms.	~9 min
Rest	Infinitely many numbers, and a few lines of algebra.	~2 min
Nutrition	Intellectual: healthy scepticism about 'always'.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Believing that agreeing at a few values proves an identity.
Ask: “Have you tried every number? Could one you haven’t tried break it?”

$(x + 3)^2 = x^2 + 9$.
Ask: “Try $x = 1$. Do the sides agree?” Then draw the grid.

$2x^2 = (2x)^2$.
Ask: “What does the square belong to in each?”

Thinking a formula and an identity are the same.
Ask: “Is it true for every value, or does it link quantities that depend on each other?”

Using a value where the sides agree as a counterexample.
Ask: “Are the two sides different at that value?”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: substitution, including negative numbers, and expanding a bracket. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner tests $(x + 3)^2$ and $x^2 + 9$ at $x = 0$, sees 9 and 9, and declares it an identity. The aim is a question that lets them reconsider.

- “What happens at $x = 1$?”
- “How many values would you need to test to be sure?”

- “Can you expand $(x + 3)^2$ to check?”
- “What does one disagreement tell you?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Is a formula an identity?
Not in this lesson's sense. A formula such as $A = \pi r^2$ links quantities that vary together; an identity says two expressions are equal for every value.

Is the proof stretch required?
It is a stretch. The secure core is: test to explore, counterexample to disprove, expand to prove.

Why are graphs shown?
They make 'for every value' visible: an identity's two sides draw the same line. The graphs use numbered, labelled axes.

What comes next?
Lesson 2: expanding and factorising, using the area model both ways.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.

Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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